CENG 4011 Senior Design Project
Final Project Report

Student Information

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Development of a User-Friendly Mobile Application for Personalized Anticoagulant Dose Optimization and Monitoring Based on INR Values

ABSTRACT

This research project aims to develop a user-friendly mobile application that automates INR (International Normalized Ratio) monitoring for patients using anticoagulant medications such as Warfarin (Coumadin/Warfarin). Due to the presence of prosthetic heart valves and other thromboembolic risks, these patients require such measurements either temporarily or on a lifelong basis. The current treatment modality typically necessitates regular monthly blood tests to monitor INR values, followed by dose adjustments made by physicians based on these results. This process entails challenges for patients, including monthly hospital visits, appointment scheduling difficulties, time consumption, and constant uncertainty. By utilizing INR results obtained from home testing or primary care centers, our application will enable patients to quickly determine their new weekly dosages through simple calculations based on clinical guidelines and expertise. The most distinctive feature of the project is that it minimizes the margin of error and enhances the accuracy of dosage recommendations by accounting for the impact of dietary intake (particularly foods rich in Vitamin K) and other medications, such as antibiotics, which may interfere with INR levels. The application is designed with simplicity and clarity (large fonts, clear visual calendars) to ensure ease of use even for elderly or non-tech-savvy users. Consequently, patients will receive immediate and reliable guidance on critical situations, such as "when to consult a physician" or "when to seek emergency care." This will enhance patient comfort, promote treatment adherence, and alleviate unnecessary burdens on the healthcare system. We plan to commercialize the mobile prototype on the App Store and Google Play in collaboration with Kedi Mobile Application Inc., a developer specializing in health data. We believe this initiative will significantly enhance the quality of life for hundreds of thousands of patients using anticoagulants.

1. SCIENTIFIC QUALITY OF THE RESEARCH PROPOSAL

1.1. Aims and Objectives

The primary objective of this research is to develop a user-friendly application prototype accessible via Android and iOS platforms. This application will automate the tracking and adjustment of weekly doses based on INR values for chronic patients undergoing treatment with Warfarin/Coumadin and similar anticoagulants. Through this prototype, we aim to provide a solution that enhances user convenience while operating in strict accordance with clinical protocols.

Problem Statement and Research Gap: It is imperative to maintain the INR values of patients on anticoagulant therapy within the target range of 2.0 to 3.5, depending on their clinical condition [1, 2, 3]. This is critical for balancing the risks of thromboembolism and hemorrhage. Under the current healthcare model, patients must undergo regular blood tests and consult their physicians for dosage escalations or reductions [6]. This process is demanding in terms of scheduling, transportation, and time. It negatively impacts the quality of life for elderly and chronic patients while creating unnecessary strain on healthcare providers. Although home INR monitoring devices are available, adjusting doses based on these results remains a challenge. The proposed project aims to address this gap. By utilizing a fundamental mathematical logic for dose adjustments, the

application will provide reliable weekly dosage recommendations based on the INR values entered by the patient. This will enable patients to manage their daily dosages from home, except in emergencies.

Concrete Objectives: The specific objectives of the project are as follows:

- **Development of a Dose Optimization Algorithm Based on Clinical Protocols:** To establish a precise backend algorithm that utilizes the patient's target INR values and historical INR patterns, following established principles for dosage adjustments.
- **Integration of an Interaction Matrix:** To incorporate user data entry for Vitamin K-containing foods and medications—including antibiotics and aspirin—that may interfere with INR levels. This will enhance the accuracy and personalization of dosage recommendations.
- **Creation of a User-Friendly Interface (UI/UX):** To design a mobile interface prototype that simplifies complex dosage regimens through the use of visual calendars. This interface will be minimalist and intuitive, ensuring accessibility for all users, including those with limited technological literacy.
- **Establishment of an Advanced Risk Management System:** To integrate a safety feature into the application that provides emergency alerts (e.g., U-1, U-4) and instructions for physician referrals if INR values fall outside critical ranges.

In collaboration with Kedi Mobile Application Inc., a developer focused on health data, the project aims to integrate backend and frontend components to deliver a testable mobile application prototype with significant commercial potential.

1.2. Innovative Aspect and Technological Value

The proposed mobile application project is innovative in that it goes beyond being a passive monitoring tool, unlike existing technological solutions and market applications in the field of Anticoagulant Dose Optimization, by offering an active and personalized Dosage Management System.

Technological Gap Analysis and Superiorities

Online dose adjustment tools such as bilektenangiyo.com [14] and mobile applications (Avalvi Care INR, INR-Coagulation, mINR, INR Diary) [15,16,17,18] help users record and visualize their dosages and target INR ranges. The fundamental deficiencies of these applications and how our project improves upon them are explained below:

- **Autonomous and Clinical Decision Support:** In all reviewed mobile applications (Avalvi Care INR, INR-Coagulation, mINR, INR Diary), dosage adjustments are left entirely to

the user, and the sole purpose is to track usage. Our project radically departs from this passive approach. The application automatically determines target INR levels based on the patient's indication for medication and automatically calculates the weekly dosage according to the entered INR value. This feature transforms our project from a simple tracking tool into an active decision support system based on clinical protocols.

- **Holistic Data Integration:** The reviewed online tools and mobile applications do not account for Vitamin K-rich foods or other medications, such as antibiotics and paracetamol, that affect warfarin metabolism and influence INR values. Our project utilizes an INR-Drug-Food Interaction Matrix (See: Appendix-3) to account for these environmental and biological variables, providing a more comprehensive and reliable personalized recommendation than existing solutions [5].

- **Comprehensive Risk Management and Active Guidance:** Contrary to the general tracking and reminder functions of current applications, our project actively warns the patient to consult a physician, skip a dose (U-4), or recognize emergency department admission symptoms (U-5) in cases of critical INR ranges or observed emergency bleeding symptoms. These safety protocols maximize the clinical reliability of our project compared to rival solutions that ignore interactions.
- **Accessibility and User-Friendly Design:** Although some reviewed applications have simple interfaces (e.g., INR Diary), our project's Figma designs (See Appendix 2) offer a unique interface innovation aimed at the error-free implementation of complex dosage (half/quarter tablet) regimens, specifically optimized for elderly users and those with low technological literacy.

Relationship with Development Plan and Strategy

This project is directly linked to the field of Health Technologies, which is a part of the 2030 Industry and Technology Strategy. The mobile prototype we will develop is a valuable output that will improve Out-of-Hospital Treatment Management for Warfarin patients and support efficiency and digital transformation in healthcare services. Furthermore, our project holds commercial potential by filling a gap in the field of active decision support, where current applications remain insufficient.

2. METHOD

The proposed mobile application project will be conducted through an interdisciplinary approach (Clinical Information Systems and Mobile Software Development) integrated with the Agile Development Methodology to achieve the specified goals and objectives. The methodology is based on scientific foundations (literature/clinical protocols) and aims for a functional mobile prototype as a tangible industrial output.

2.1. Development of the Dose Optimization Algorithm

This stage constitutes the most critical and analytical solution method of the project.

- **Clinical Data Collection and Standardization:** Protocols from established clinical literature (e.g., INR Value – Dose Proportioning) that form the basis of Warfarin dosage adjustment will be examined. The boundaries of the algorithm will be determined based on target INR values (e.g., between 2.5–3.5 for prosthetic heart valves) [7].
- **Algorithm Design:** A mathematical model will be designed to calculate how much the current dose should be increased or decreased according to the patient's target INR value. The model will be based on simple linear regression principles or formulas commonly used in literature that rely on the relationship between the patient's total weekly dose and the current INR value [4].

Representative Patient Data and Mathematical Calculations:

a. A patient who underwent Mitral valve prosthesis surgery 3 years ago and has been using a full dose of "Warfarin 5 mg tablet" daily for the last month (total weekly dose 35 mg) has a new INR measurement of 1.8.
    - The weekly dose is increased by 1/7 to 40 mg, distributed across days such as 1.5 tablets 2 days a week (e.g., Monday and Thursday 1.5 tablets, other days 1 tablet).
    - Additionally, a U-1 alert is issued (physician contact required) as an additional anticoagulant injection (Enoxaparin) may be necessary to prevent clot formation on the prosthetic valve.
b. A patient using Warfmadin 5 mg for Atrial fibrillation for 4 months at a dose of half a tablet 3 days and a full tablet 4 days a week (total weekly dose 27.5 mg) has a new INR measurement of 4.2.
    - One dose is skipped, and the total weekly dose is reduced by approximately 1/7; the new dose is distributed as half a tablet 5 days and a full tablet 2 days a week.
    - Since the value is above the therapeutic target, U-3 and U-5 alerts are issued, directing the patient to a physician for potential bleeding suspicion.
c. A patient who underwent double valve prosthesis surgery 12 years ago has been taking a quarter tablet of Warfmadin 5 mg daily for the last month.
    - The new INR value is 3.2. As the current dose is within the therapeutic target range, continuation of the same dose and a follow-up after 1 month are recommended.
d. A patient using Warfmadin protectively for 1 year due to recurrent pulmonary embolism has been taking 1.5 tablets 4 days and 2 tablets the other days (total weekly dose 45 mg) of Warfmadin 5 mg for the last month.
    - The new INR value is 6.8. The patient is instructed to pause the medication for 3 days and repeat the INR test on the 4th day. U-4, U-5, and U-6 alerts are issued, with recommendations for physician referral where necessary.
e. A patient who started Warfmadin 5 mg 6 months ago due to an embolic ischemic cerebrovascular event and has been using 1 tablet 6 days and half a tablet 1 day a week (total weekly dose 32.5 mg) for the last month has a new INR measurement of 2.1.

    - The total weekly dose is increased by approximately 1/14 to 35 mg, and daily use of 1 tablet is recommended (U-2 alert).

- **Development of the Interaction Matrix:** Foods and drugs affecting INR will be listed, and a Rule-Based System will be developed to modify the algorithm's dose adjustment coefficient according to their known effects on Warfarin (increasing or decreasing the effect). This adds depth to the simple mathematical solution [8].

2.2. Mobile Application Development and User Interface (UI/UX)

- **Frontend Development:** Based on Figma designs (See: Appendix-2), the application interface will be coded using a cross-platform technology (e.g., Flutter or React Native) compatible with both Android and iOS.
- **Backend and Database Architecture (Serverless):**
    - **Database:** INR history, dosage data, and patient-specific clinical matrices (INR Interactions, Warning Protocols) forming the backbone of the project will be securely stored on Google Cloud Firestore (Firebase).
    - **RESTful API Development:** A RESTful API architecture will be used to securely process INR, nutritional, and new medication data from the mobile app and send the calculated dosage recommendation back. This API will be developed using Firebase Cloud Functions (Node.js), ensuring the algorithm runs in a reliable and scalable environment without server management.
    - **Authentication and Security:** User and data security will be provided through Firebase's built-in Authentication module and encrypted API calls via HTTPS. Sensitive health data will be stored in an anonymized format (KVKK/GDPR compliance).

2.3. Risk Management and Validation Protocols

- **Implementation of Security Mechanisms:** Emergency warning screens will be integrated into the application to automatically alert the patient and direct them to a physician/emergency room in cases where INR values drop to critical levels (clotting risk - U-1, U-2) or rise (bleeding risk - U-3, U-4). This clarifies the application's scope of responsibility.
- **Usability Satisfaction Tests and Ethical Process:** Ethical permission will be required for the critical stages of clinical validation and user satisfaction testing.
    - **Ethics Committee Application:** Applications will be made to the Muğla University Ethics Committee to test algorithm accuracy with real patient data and conduct usability satisfaction tests.
    - **Accuracy Testing:** Following ethical approval, under the supervision of Assoc. Prof. Dr. Zekeriya Arslan (Medical/Industry Advisor), the algorithm's accuracy will be tested by comparing the dosages recommended by the application against reference dosages determined by clinicians using anonymized historical and current INR data from 10 volunteer patients.
    - **Satisfaction Testing:** In addition to clinical tests, user interaction flows and interface ease-of-use will be quantitatively evaluated through standardized surveys such as the SUS (System Usability Scale).

3. PROJECT MANAGEMENT

3.1. Work Schedule

Date Range	Activities	By Whom It Will Be Performed	Success Criteria, and Contribution to Project Success
01 / 02 / 2026 – 15 / 04 / 2026	**Planning, Interface Coding, and Backend Infrastructure Preparation:** Final confirmation of the project's clinical protocols. Commencement of cross-platform interface coding (e.g., Flutter/React Native) based on Figma designs (See: Appendix-2). Design of the Google Cloud Firestore database schema.	Researcher	Completion of fundamental interface flows (Login, Personal Information). Implementation of the Firebase Cloud Firestore setup.
16 / 04 / 2026 – 31 / 06 / 2026	**Backend and Database Integration:** Establishing the foundation of the RESTful API (Firebase Cloud Functions) to ensure secure data communication between the algorithm and the mobile interface. Uploading the Interaction Matrix (See: Appendix-3) data to Firestore.	Researcher	Establishment of the RESTful API architecture for algorithmic calculations. Ensuring successful data exchange between the mobile application and the database.
01 / 07 / 2026 – 31 / 08 / 2026	**Algorithm and Core Development:** Coding of the Dose Optimization Algorithm. Ensuring the algorithm returns the correct dose recommendation by utilizing INR and interaction data transmitted via the API.	Researcher	The Dose Optimization Algorithm reaching a testable state. The algorithm operating with 99% accuracy via simulation data within the mobile application.
01 / 09 / 2026 – 31 / 10 / 2026	**Core Integration and Functionality:** Integration of complex interface elements, such as the Weekly Dosage Calendar (See: Appendix-2), with the algorithm output. Completion of User Authentication and fundamental security modules.	Researcher	The application reaching a functional end-to-end Beta version. Secure operation of the User Authentication system.

01 / 11 / 2026 – 31 / 11 / 2026	**Risk Management and Alert Coding:** Coding of all Risk Management Protocols (from U-1 to U-12). Implementation of triggers for active emergency alerts (U-5) during critical INR entries.	Researcher	Full integration of the Risk Management module and the critical alert system functioning with 99% accuracy.
01 / 12 / 2026 – 31 / 12 / 2026	**Testing and Ethics Committee Application:** Comprehensive internal (End-to-End) testing and debugging of the application. Preparation of the application file for the Muğla University Ethics Committee regarding clinical accuracy and satisfaction tests.	Researcher, Academic Advisor	The application reaching its final version readiness. Submission of the Ethics Committee application.
01 / 01 / 2027 – 31 / 01 / 2027	**Ethical Approval, Clinical Accuracy Tests, and Reporting:** Conducting algorithm accuracy and user satisfaction tests with 10 volunteer patients following Ethics Committee approval. Preparation and submission of the Project Final Report.	Researcher, Academic Advisor	Confirmation of the algorithm accuracy test by the clinician. Complete submission of the Project Report and preparation of academic output (draft of a paper/presentation).

3.2. Risk Management

Most Important Risks	Measures To Be Taken (Plan B)
Risk 1: High Error Margin in Algorithm Output (Clinical Reliability Risk) Due to the patient's individual metabolism and numerous external factors (undisclosed drug/food interactions), the developed dose adjustment algorithm may exhibit a high margin of error in achieving the expected INR value with the recommended weekly dosage. This situation poses a critical risk to patient health.	**Plan B:** Algorithm Limitation and Professional Approval Mechanism **Measure A**: Restricting the use of the algorithm to stable patients without complications; activating immediate physician referral warnings (U-5, U-8) for complex cases (e.g., persistent INR values outside the target range). **Measure B:** Subjecting all algorithm-generated calculations to additional validation tests using anonymized real case scenarios by a medical consultant (Assoc. Prof. Dr. Zekeriya Arslan) before presenting any dose recommendations.
Risk 2: Difficulty of Use by the Target Population (UI/UX and	**Plan B:** Interface Simplification Revision and Support Modules

Adoption Risk) Despite a simple design, elderly patients and those with limited technological familiarity—who constitute the target population of the application—may experience difficulty using the application or may enter vital data (INR value, newly introduced medications) incorrectly. This may render the application output meaningless.	**Measure A:** Developing simplified interface features based on user feedback, adopting a "One-Click" usage principle. **Measure B:** Integrating simple educational modules into the in-app "Help" section (see: Figma Menu Screen), where each step is explained visually and audibly. Additionally, providing informative content on topics such as how INR values are measured.
Risk 3: Technical/Personnel Changes in the Industrial Partner Changes in technical consultants or personnel within the collaborating Industrial Partner during the project period may negatively affect the timely completion of the project and its commercialization objectives.	**Plan B:** Open-Source Code Documentation and New Consultant Integration **Measure A:** Preparing detailed and up-to-date documentation of the project's software architecture and algorithm details (including the backend) from the outset, thereby reducing the onboarding time for new consultants or team members. **Measure B:** Identifying a secondary mentorship resource from within the university (in coordination with the Project Coordinator's Academic Advisor), preferably from the field of software engineering, to ensure technical continuity in the event of potential disruptions.

3.3. Research Opportunities

Institution Hosting the Infrastructure/Equipment	Type and Model of the Available Infrastructure/Equipment within the Institution (e.g., laboratory, vehicle, machinery/equipment, etc.)	Purpose of Use within the Project
Mugla Sitki Kocman University	Süzek Lab Servers	In this project, the facilities of the Süzek Laboratory (SuzekLab) located at the Muğla University Research Laboratories Center will be utilized. The laboratory provides a dedicated office space for undergraduate students involved in project management, as well as 11 high-performance servers

		(Dell PowerEdge R530 with 128–256 GB RAM, 8 TB storage, Ubuntu 20.04). The software will be developed on these high-performance Linux servers. The TCGAnalyzeR back-end files will be hosted on these servers, and cohort and drug response datasets will be downloaded from them.
Mugla Sitki Kocman University	Kütüphane	In addition, the university's library resources will be used to access literature available in both printed and online formats.

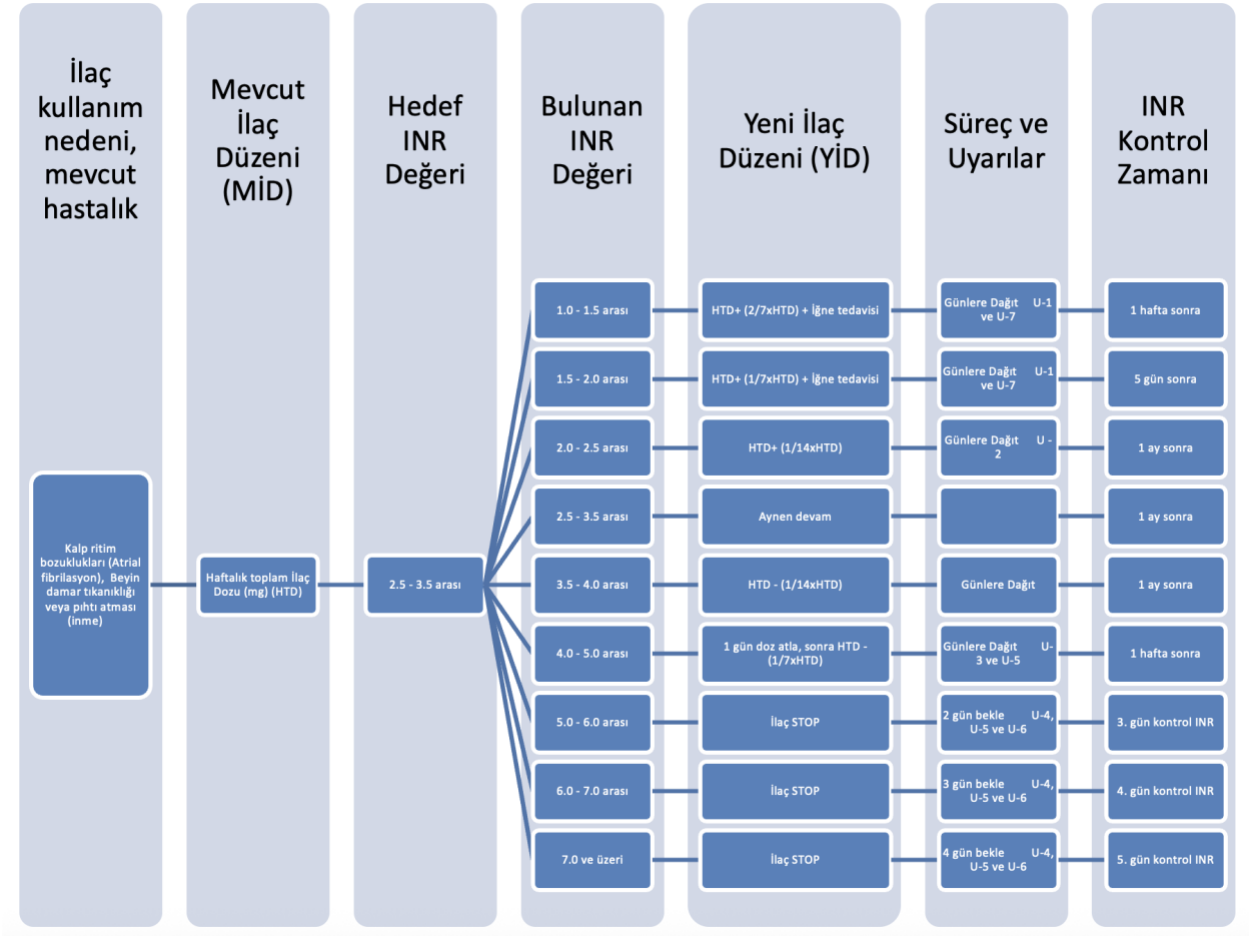
4. INDUSTRY-FOCUSED OUTPUTS AND WIDESPREAD IMPACT

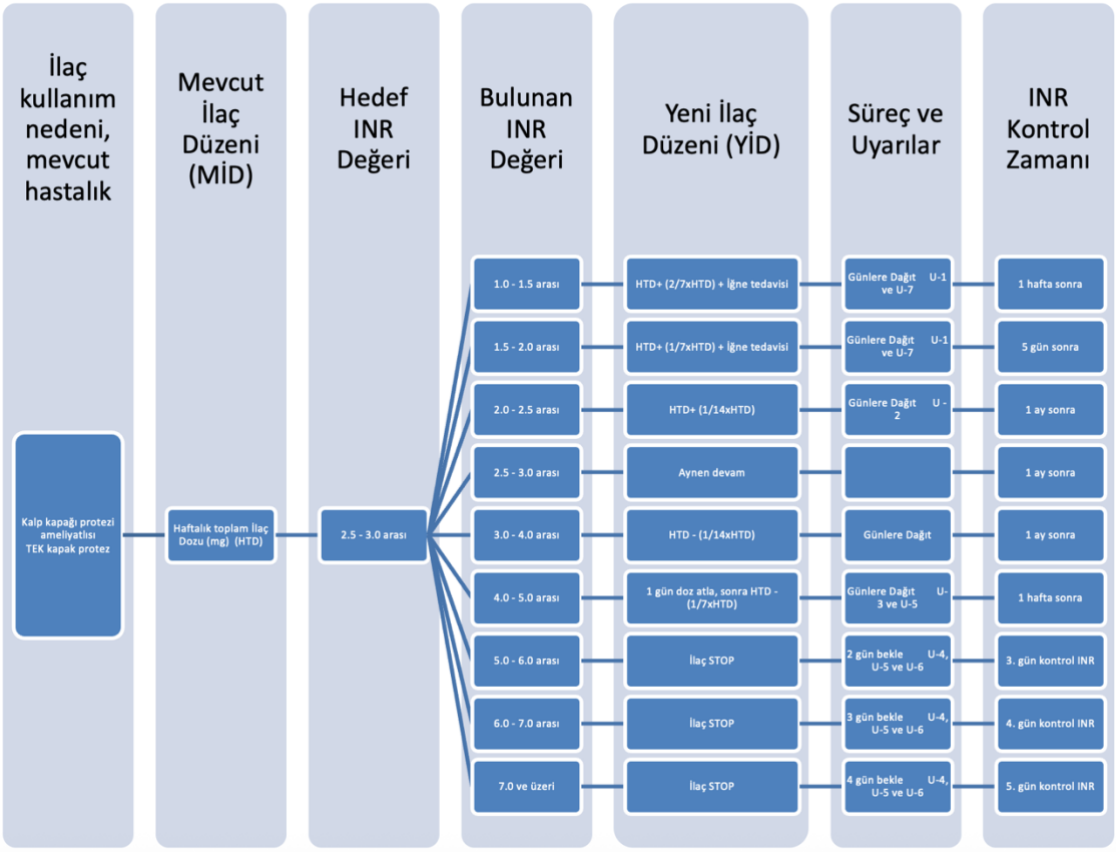
Outputs, Impact, and Outcomes	Anticipated Output(s), Impact(s), and Outcome(s)
Scientific/Academic Outputs (National/International articles, book chapters, books, conference papers, etc.)	Within the scope of this study, it is aimed to present an academic article on the use of technology in the medical field based on the results of the undergraduate graduation project conducted. In addition, it is planned to investigate how the relevant topic can be further advanced through artificial intelligence.
Economic/Commercial/Social Outputs (Products, prototypes, patents, utility models, registrations, visual/audio archives, inventories/databases, workshops, training activities, scientific events, etc.)	The contribution of our project to healthcare institutions lies in transferring INR dosage adjustment—which requires continuous physician supervision and thus causes persistent fatigue for patients and congestion in hospitals—into an individualized and optimized application. In this way, the process will be simplified for the patients concerned, and the workload of physicians will be reduced. The application is planned to be launched primarily in Türkiye. However, after obtaining the necessary patents, it is aimed to add features for international use and enable global deployment, thereby providing benefits worldwide and establishing recognition in this field.
Outputs Aimed at Generating New Project(s) (New national/international projects, etc.)	Within the scope of the project, feasibility studies are planned to adapt the algorithm and risk management model to the monitoring of other chronic diseases such as diabetes and hypertension [9,10]. Based on the developed prototype and business model, it is aimed to apply to the TÜBİTAK BİGG program in order to maximize the commercialization potential.

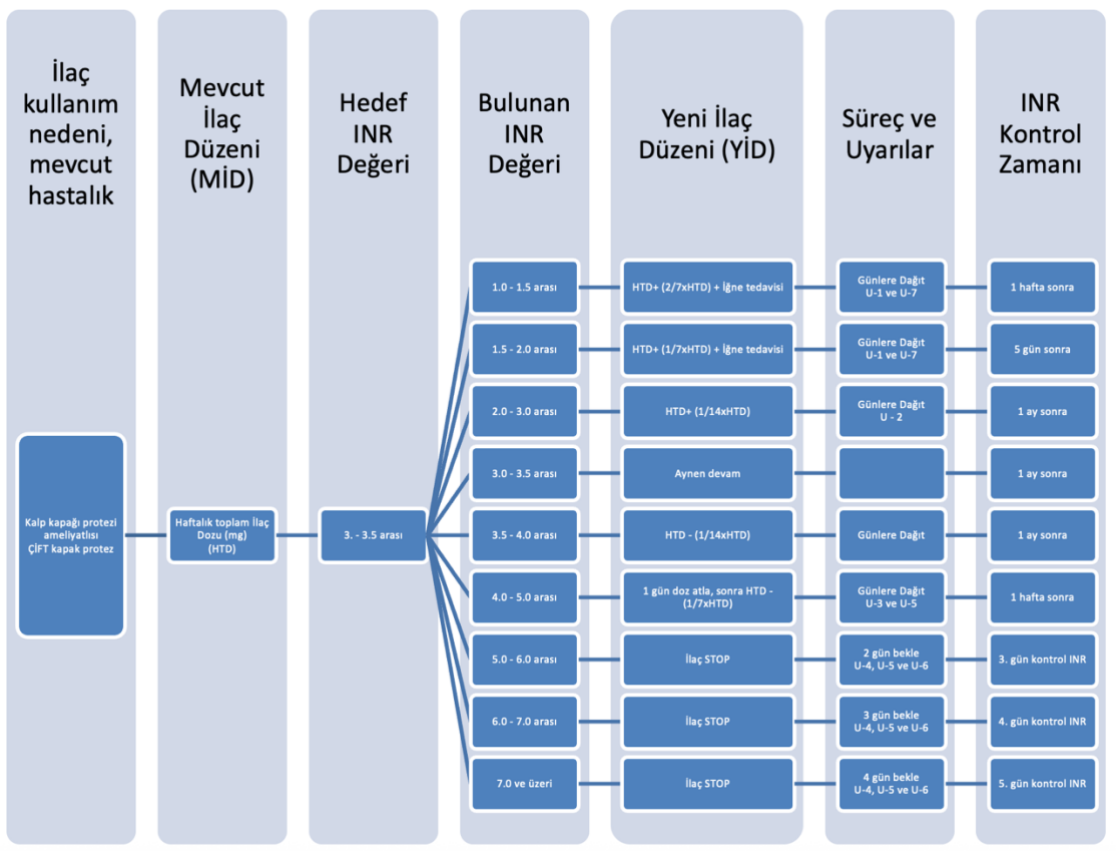
5. Other Issues To Specify

The student prepared for the project by completing the courses Mobile Application Development, Database Management Systems, and Applied Machine Learning in the third year, as well as Modeling and Simulation, Data Science, and Analytics, in the Fall 2025 semester.

The Dose Adjustment Algorithm and Risk Management Protocol, which form the foundation of the project, are concretized through the schematic presented below. The application determines the new dosage regimen (New Dosage Instruction, NDI) and the next control time based on the patient's current indication for medication use (e.g., cardiac arrhythmia), target INR range (e.g., 2.5–3.5), and the measured INR value. This schematic clearly illustrates the application's mechanisms for discontinuing medication (Medication STOP) and directing the patient to emergency care (U-5) in critical situations (e.g., INR values above 4.0). This protocol ensures the clinical reliability of the project and clearly defines its scope of responsibility.




İlaç kullanım nedeni, mevcut hastalık
Mevcut İlaç Düzeni (MİD)
Hedef INR Değeri
Bulunan INR Değeri
Yeni İlaç Düzeni (YİD)
Süreç ve Uyarılar
INR Kontrol Zamanı
Kalp kapağı protezi ameliyatlısı TEK kapak protez
Haftalık toplam İlaç Dozu (mg) (HTD)
2.5 - 3.0 arası
1.0 - 1.5 arası
HTD+ (2/7xHTD) + İğne tedavisi
Günlere Dağıt U-1 ve U-7
1 hafta sonra
1.5 - 2.0 arası
HTD+ (1/7xHTD) + İğne tedavisi
Günlere Dağıt U-1 ve U-7
5 gün sonra
2.0 - 2.5 arası
HTD+ (1/14xHTD)
Günlere Dağıt U - 2
1 ay sonra
2.5 - 3.0 arası
Aynen devam
1 ay sonra
3.0 - 4.0 arası
HTD - (1/14xHTD)
Günlere Dağıt
1 ay sonra
4.0 - 5.0 arası
1 gün doz atla, sonra HTD - (1/7xHTD)
Günlere Dağıt U-3 ve U-5
1 hafta sonra
5.0 - 6.0 arası
İlaç STOP
2 gün bekle U-4, U-5 ve U-6
3. gün kontrol INR
6.0 - 7.0 arası
İlaç STOP
3 gün bekle U-4, U-5 ve U-6
4. gün kontrol INR
7.0 ve üzeri
İlaç STOP
4 gün bekle U-4, U-5 ve U-6
5. gün kontrol INR


İlaç kullanım nedeni, mevcut hastalık
Mevcut İlaç Düzeni (MİD)
Hedef INR Değeri
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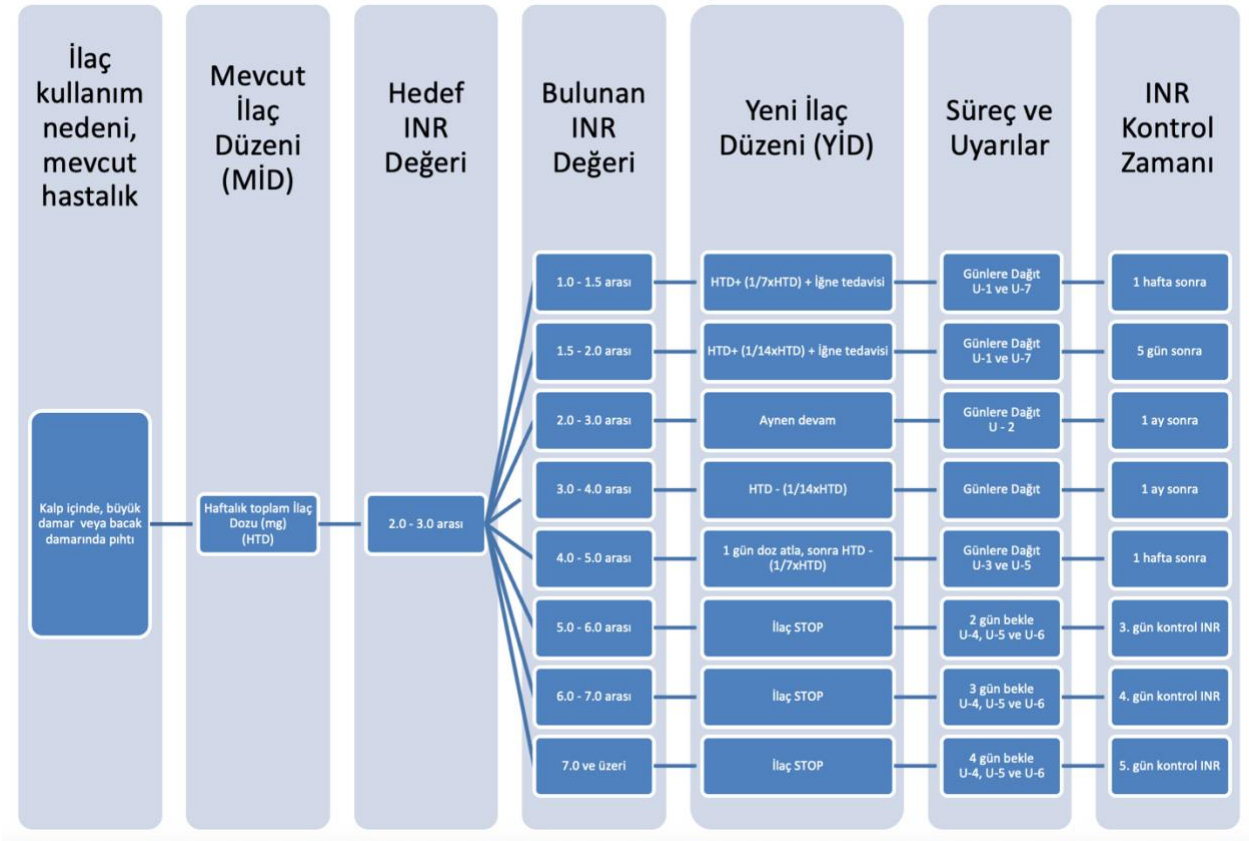
Yeni İlaç Düzeni (YİD)
Süreç ve Uyarılar
INR Kontrol Zamanı
Kalp kapağı protezi ameliyatlısı ÇİFT kapak protez
Haftalık toplam İlaç Dozu (mg) (HTD)
3. - 3.5 arası
1.0 - 1.5 arası
HTD+ (2/7xHTD) + İğne tedavisi
Günlere Dağıt U-1 ve U-7
1 hafta sonra
1.5 - 2.0 arası
HTD+ (1/7xHTD) + İğne tedavisi
Günlere Dağıt U-1 ve U-7
5 gün sonra
2.0 - 3.0 arası
HTD+ (1/14xHTD)
Günlere Dağıt U - 2
1 ay sonra
3.0 - 3.5 arası
Aynen devam
1 ay sonra
3.5 - 4.0 arası
HTD - (1/14xHTD)
Günlere Dağıt
1 ay sonra
4.0 - 5.0 arası
1 gün doz atla, sonra HTD - (1/7xHTD)
Günlere Dağıt U-3 ve U-5
1 hafta sonra
5.0 - 6.0 arası
İlaç STOP
2 gün bekle U-4, U-5 ve U-6
3. gün kontrol INR
6.0 - 7.0 arası
İlaç STOP
3 gün bekle U-4, U-5 ve U-6
4. gün kontrol INR
7.0 ve üzeri
İlaç STOP
4 gün bekle U-4, U-5 ve U-6
5. gün kontrol INR



6. Appendices

Appendix-1: Resources

[1] Joglar, J, Chung, M, Armbruster, A., et al. 2023 ACC/AHA/ACCP/HRS Guideline for the Diagnosis and Management of Atrial Fibrillation: A Report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. JACC. 2024 Jan, 83 (1) 109–279
https://www.jacc.org/doi/abs/10.1016/j.jacc.2023.08.017

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https://academic.oup.com/eurheartj/article/45/36/3314/7738779

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https://www.ahajournals.org/doi/10.1161/CIR.0000000000001193

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[13] Warfarin (Coumadin) İlaç ve Besin Etkileşimleri Tıbbi Referans Kılavuzu

Web Tool:

[14] Bilekten Anjiyo (2025). Warfarin Doz Ayarlama (INR Değerine Göre). Erişim adresi: https://www.bilektenanjiyo.com/warfarin/warfarin-dose-adjustment.php

Mobile application analytics:

[15] Avalvi Care INR Mobil Uygulama (App Store)

[16] INR Diary Mobil Uygulama (Google Play Store / App Store)

[17] mINR (App Store)

[18] INR-Coagulation-Manager (AppStore)

Appendix-2: Mobile Application Figma Design Outputs

The key screen visuals that serve as concrete evidence of the user-centric and accessible interface (UI/UX) emphasized in the *Innovative Aspect of the Project (Section 1.2)* and *Methodology (Section 2)* are presented in this annex.

These designs illustrate the critical workflows of the application:

- **Login and Personalization:** Following the Login and Registration screens, the workflow demonstrates the collection of treatment-critical information such as the patient's Personal Information (age, gender) and Existing Comorbidities (e.g., Atrial Fibrillation, Prosthetic Heart Valve).
- **Weekly Dosage Visualization:** The Weekly Medication Dosage Calendar on the Home Screen aims to visually simplify complex dosage regimens (e.g., half/quarter tablets).
- **Interaction Management:** The lists of Drugs and Foods Affecting INR (e.g., spinach, cabbage, antibiotics), accessible via the menu, concretely demonstrate the application's holistic and personalized dose recommendation capability.

#1

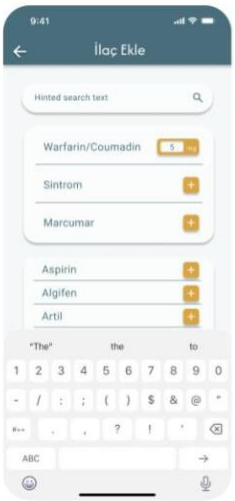
#2

#3

#4

#5

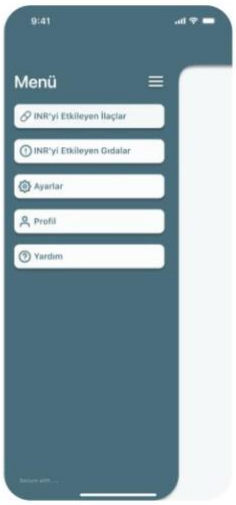
İlaç Ekle



INR Değeri

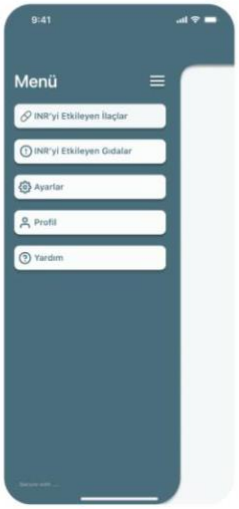
Anasayfa

Menü

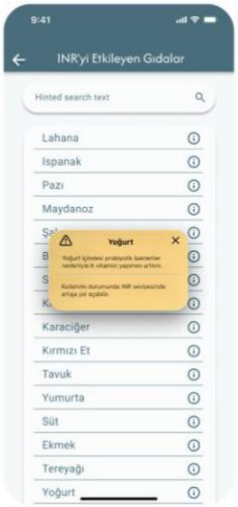


Profil

Menü



Dikkat Edilmesi Gerek...



Dikkat Edilmesi Gerek...

Appendix-3: Interaction Matrix (Drug/Food Details)

This annex presents detailed lists of external factors that are clinically known to affect the metabolism of Warfarin, and that form the basis of the project's **Scientific Merit (Section 1.1)** and the **Dose Optimization Algorithm** (see: *Section 2 – Methodology*). The application provides personalized management by incorporating these user-entered factors into the dose recommendation process.

These lists include:

- **Foods Affecting INR:**
  A distinction is made between foods rich in or dense with vitamin K that reduce INR values and thus diminish the effect of Warfarin (e.g., cabbage, spinach, parsley, liver, broccoli, green tea, avocado, etc.) and foods that have a lesser effect on Warfarin activity (e.g., red meat, chicken, eggs, tomatoes, rice, etc.). Yogurt is noted to increase vitamin K production due to its probiotic content.
- **Drugs Directly Affecting the Coagulation System:**
  This list includes drug groups that directly affect the coagulation system—such as aspirin, ibuprofen, ketoprofen, and naproxen derivatives—which generally should not be used concomitantly with Warfarin without physician approval.
- **Drugs Increasing the Effect of Warfarin:**
  Factors that increase INR levels and thereby elevate the risk of bleeding (e.g., alcohol, antibiotics, amiodarone, paracetamol, corticosteroids, vitamin E, thyroid hormone, etc.) are specified in detail.
- **Drugs Decreasing the Effect of Warfarin:**
  Factors that reduce INR levels and thereby increase the risk of thrombosis (e.g., antithyroid drugs, oral contraceptives, carbamazepine, ascorbic acid, etc.) are indicated.

This detailed matrix illustrates the scientific rigor underlying the application and its capability for **Personalized Dose Optimization**. In the presence of these medications, warning protocols will be activated.

Appendix-4: Detailed Warning Texts Protocol

All warning and guidance texts (from U-1 to U-12), prepared based on clinical data and prioritizing patient safety, which support the approach outlined in the **Risk Management (Section 3.1)** and **Methodology (Section 2)** sections of the project, are presented in detail in this annex. These protocols demonstrate how the application not only provides dosage recommendations in critical situations but also fulfills its responsibility to protect patient health.

The warnings mainly cover the following categories:

- **Emergency and Critical Risk Warnings:**

- **Risk of Thrombosis (U-1, U-2):** Warnings indicating the potential need for subcutaneous injections when the INR value is very low, along with the specification of a weekly regimen aimed at increasing the dose.
- **Risk of Bleeding (U-3, U-4):** Instructions for temporarily discontinuing the medication and determining a new control time when the INR value is high or very high.
- **Life-Saving Emergency Referral (U-5):** A critical warning text directing the patient immediately to the emergency department in the presence of any life-threatening bleeding symptoms, such as vomiting blood, black stools, severe headache, or extensive bruising.

- **Treatment Adherence and Educational Warnings:**
  - **Dosing and Monitoring Instructions (U-8, U-9):** Clear instructions regarding the timing of medication intake and when the next INR test should be performed.
  - **Missed Dose Management (U-12):** Explicit guidance on when a double dose should not be taken in case of a missed dose and how treatment should be continued.
  - **Lifestyle Management (U-6, U-7):** Recommendations on restricting or increasing the intake of vitamin K–rich foods depending on the INR value.
- **Medical Intervention Warnings (U-10, U-11):** Preventive and informative warnings advising the patient to inform healthcare providers about Warfarin use before any interventional procedure and to carry a Warfarin card in their wallet for emergencies.

These protocols were prepared under the supervision of the academic advisor and are based on detailed and clinically accepted guidelines.